

DRM walk — every screen, step by step

Before you start — five minutes, once

- 1. On the shop PC, double-click **START.cmd**; it prints the address for the tablet.
- 2. On the tablet, turn **mobile data off** and join the shop Wi-Fi.
- 3. Install the certificate on the tablet once — **server\certs\lan-cert.crt**, as in TABLET_TRUST.md — so the address opens with no warning and the camera and offline mode are allowed.
- 4. Open the address START.cmd printed and accept *Add to home screen*.
- 5. Sign in with the DRM account.

Then go step by step. Each step shows the screen as it should look, the taps in order, what must be on your screen, and what to suspect when it is not. Tick **Seen it** only when you saw it.

Verified on the packaged build

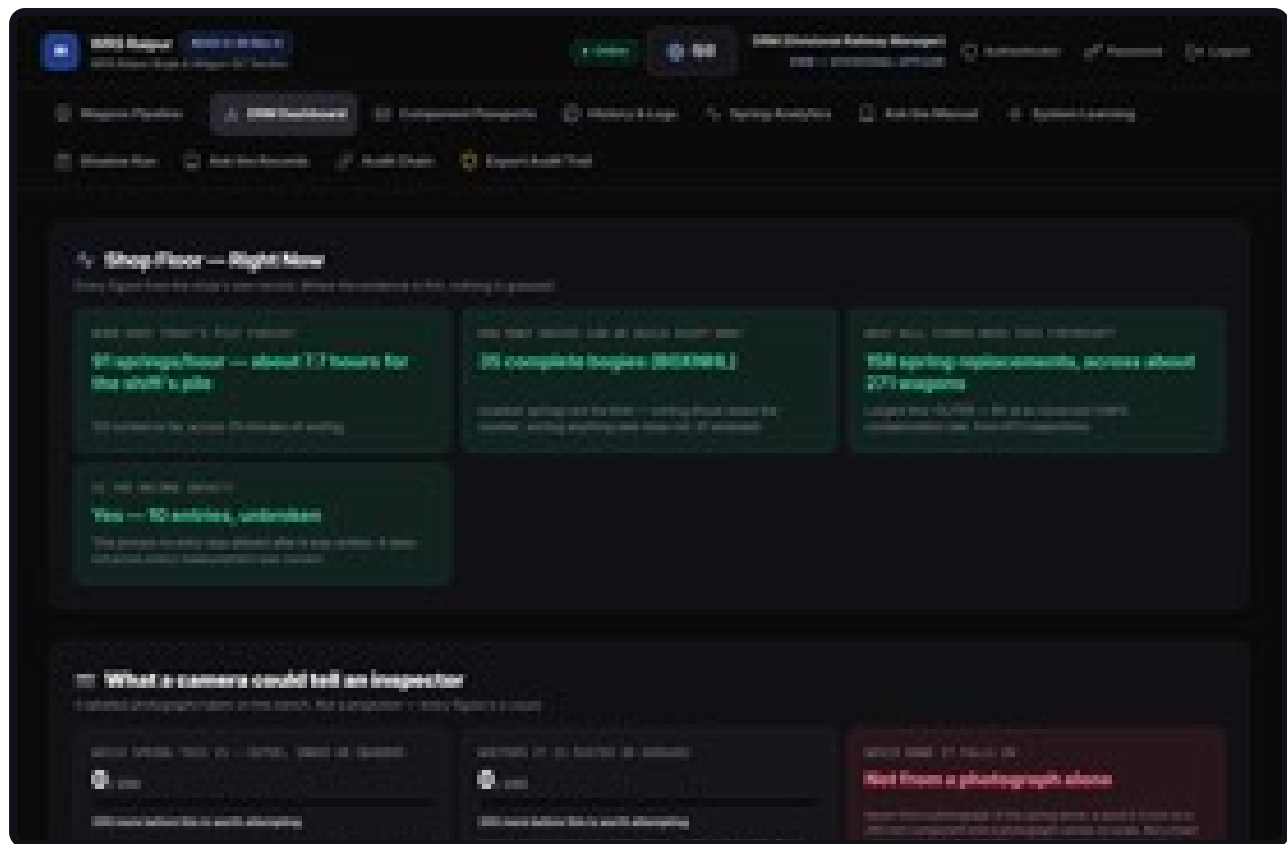
9 / 9 steps verified

When 21 September 2026
On the packaged bundle, seeded with the demonstration record, on a laptop screen
By hand 1 step (4.9) need a person or a second device

- 4.1 – 4.4 · The dashboards
- 4.5 – 4.9 · Oversight — what the DRM can see and cannot do

The dashboards 4.1 – 4.4

Sign in as the DRM. Every figure from the record, with its count.



dashboard

DO

1. Sign in with the account you were given.
2. Tap **DRM Dashboard**. Read the tiles under **Shop Floor — Right Now**.

YOU MUST SEE

- *When does today's pile finish?* — springs per hour and the hours for the shift's pile, with how many sorted over how many minutes.
- *How many bogies can we build right now?* — the number, and which spring position is the limit.
- *What will Stores need this fortnight?* — replacements across so many wagons, with the observed condemnation rate and the number of inspections it rests on.
- *Is the record intact?* — yes, N entries, unbroken.
- None says *Not yet known*.

WHAT IT PROVES

Four questions a DRM actually asks, answered from the record with the basis under each. Where the evidence would be thin the tile says "not yet known" instead of a number — on the demo record it is not thin.

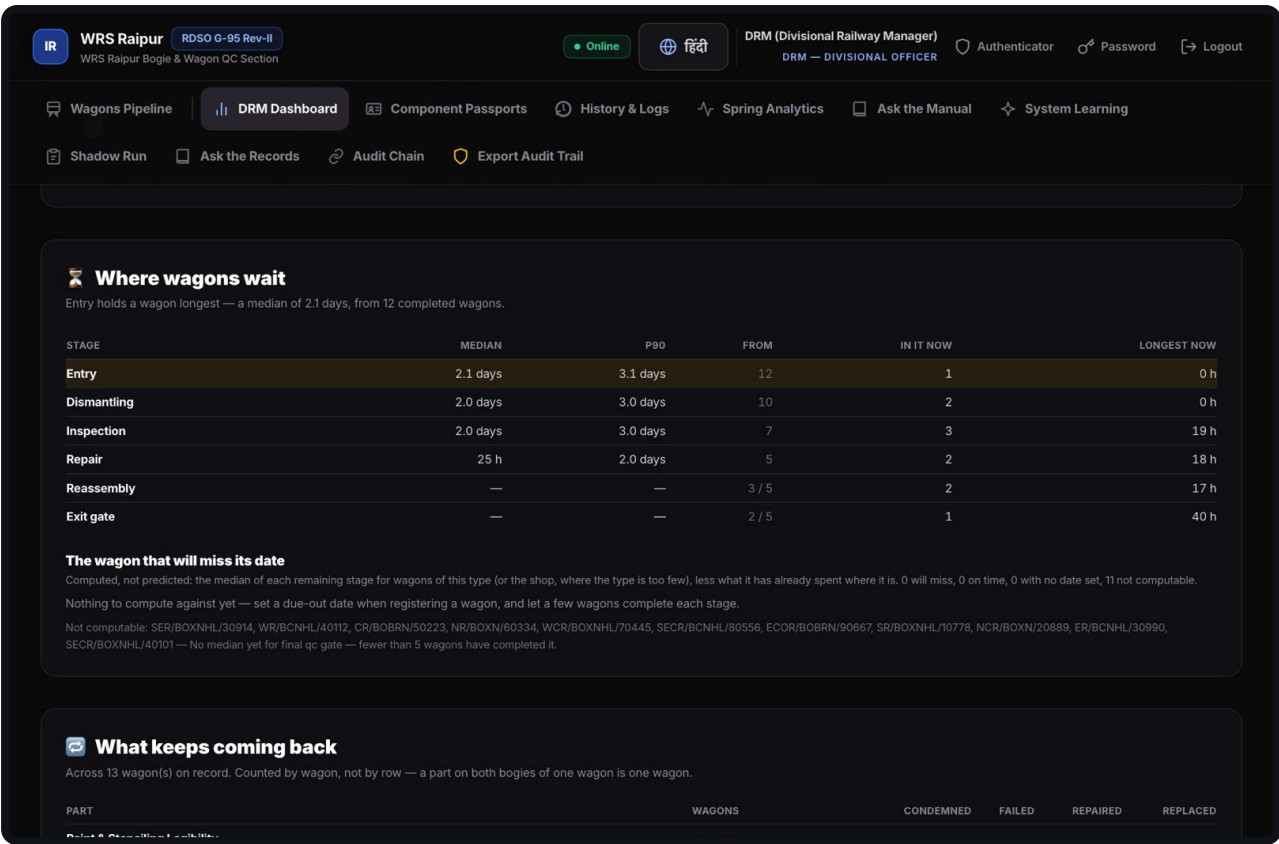
CHECKED ON THE BUILD

verified • 02:29:06 pm

- ✓ the "Shop Floor — Right Now" section
- ✓ when today's pile finishes, with the rate
- ✓ how many bogies can be built
- ✓ what Stores will need
- ✓ none says "Not yet known"

IF IT IS NOT THERE

- A tile says "Not yet known": the bench has too few springs today — sort a few as the inspector first, or re-seed.



dashboard lower

DO

- 1. Scroll.

YOU MUST SEE

- **What a camera could tell an inspector** — blind-read agreement, assembly coverage, the pocket-count dataset: what has been earned and what has not.
- **Where wagons wait** — dwell per stage with the median and the slow stage; **the wagon that will miss its date**, computed from the stages left, not predicted.
- **What keeps coming back** — recurring findings by part and wagon type.
- Turnaround, the band distribution (six colours, the bench included), inspector quality, gauge exposure — every rate with its count.

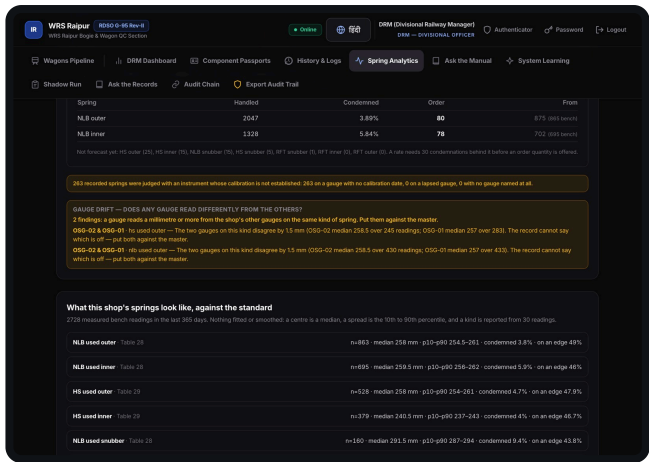
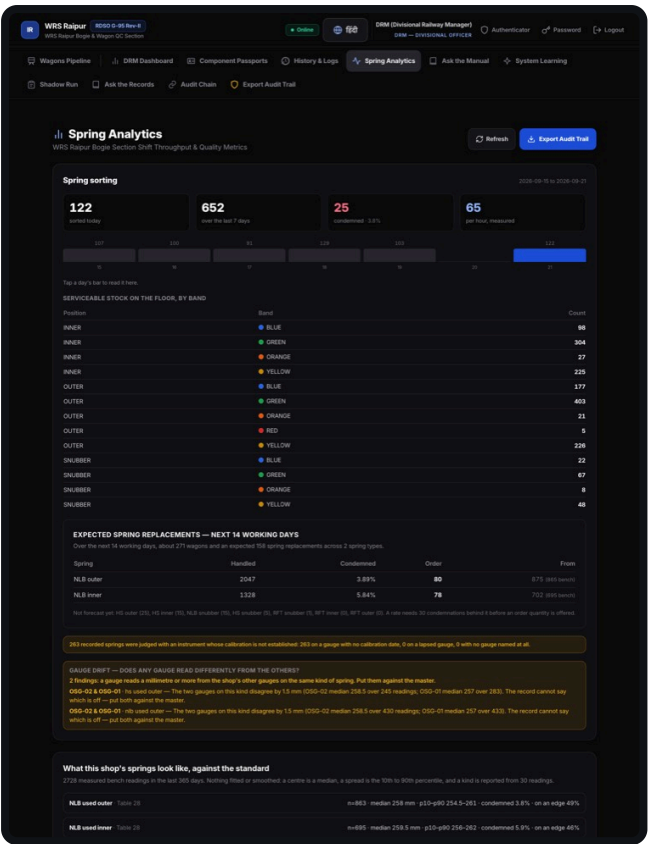
WHAT IT PROVES

Every figure carries its n. Nothing here is a forecast or a model's number.

CHECKED ON THE BUILD

- ✓ turnaround
- ✓ where wagons wait – and the wagon that will miss its date
- ✓ what keeps coming back
- ✓ inspector quality
- ✓ gauge exposure
- ✓ rates carry their counts

verified • 02:29:12 pm



analytics drift

analytics

DO

1. **Spring Analytics.** Tap one of the day bars.
2. Scroll to the standard report and the gauge drift panel.

YOU MUST SEE

- ▶ Bench totals today and this week; tapping a bar reads that day's figures beneath it.
- ▶ Stock by band.
- ▶ **The standard report** — this shop's springs against the G-95 bands, with *n* on every figure.
- ▶ **Gauge drift** names **OSG-02**: the two outer gauges disagree by 1.5 mm — put both against the master.

WHAT IT PROVES

The distribution of a shop's springs against the standard is evidence RDSO has never had from a shop; the drift check finds a gauge reading high from the record alone.

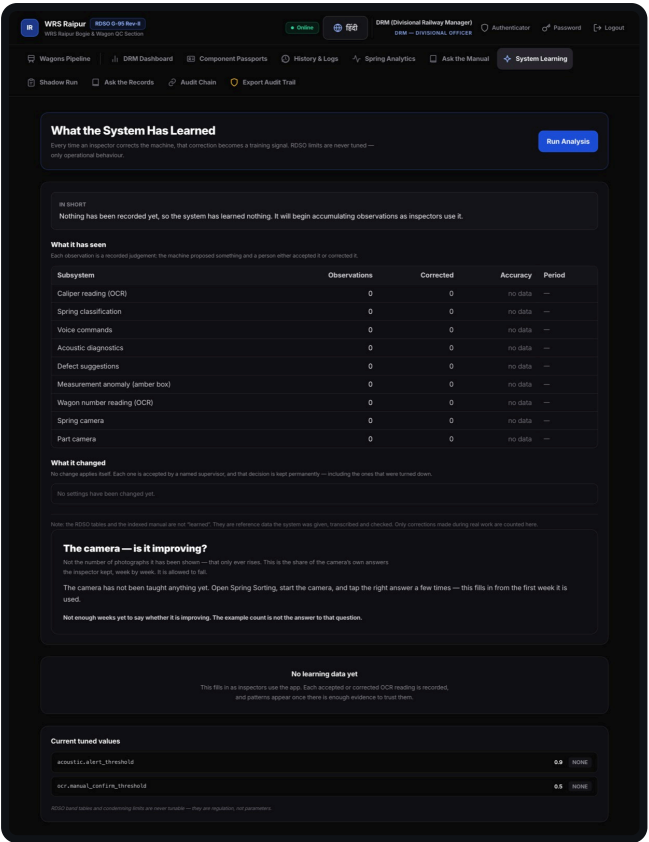
CHECKED ON THE BUILD

verified • 02:29:13 pm

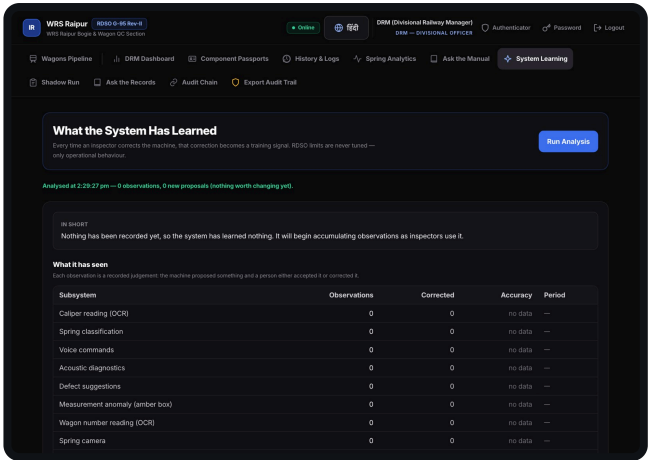
- ✓ bench totals today and this week
- ✓ tapping a day's bar reads that day's figures beneath
- ✓ stock by band
- ✓ the standard report
- ✓ n on the figures (6 found)
- ✓ the Gauge drift panel names OSG-02: "GAUGE DRIFT — DOES ANY GAUGE READ DIFFERENTLY FROM THE OTHERS? 2 findings: a gauge reads a millimetre or more from the s..."

4.4 What the system has learned

☐ Seen it



learning



learning ran

DO

1. **System Learning.** Read the table. Tap **Run Analysis**.

YOU MUST SEE

- Each subsystem — caliper reading, spring classification, voice, acoustic, defect suggestions, the amber box, wagon-number reading, spring camera, part camera — with observations, corrections and accuracy. On the demo record: *no data*.
- *The camera has not been taught anything yet.* The server confirms the camera may not sort on its own.
- After **Run Analysis**, a green line: *Analysed at HH:MM — N observations, 0 new proposals (nothing worth changing yet).*
- *RDSO band tables and condemning limits are never tunable — they are regulation, not parameters.*

WHAT IT PROVES

What has been learned is a table with counts, not a claim. No change applies itself; each is accepted by a named supervisor. The standard is never "learned".

CHECKED ON THE BUILD

verified · 02:29:21 pm

- ✓ each subsystem with its observations, corrections and accuracy
- ✓ the camera says it has not been taught anything yet
- ✓ the server confirms the camera may not sort on its own (allowed: ?)
- ✓ after Run Analysis: "Analysed at 2:29:27 pm — 0 observations, 0 new proposals (nothing worth changing yet)."

IF IT IS NOT THERE

- Run Analysis shows an error: the server refused the DRM — report it; it was fixed on 21 Sep.

Oversight — what the DRM can see and cannot do 4.5 – 4.9

Read everything; write nothing.

4.5 Shadow Run — read-only

☐ Seen it

IR

WRS Raipur

RDSO G-95 Rev-II

WRS Raipur Bogie & Wagon QC Section

Online

DRM (Divisional Railway Manager)

DRM — DIVISIONAL OFFICER

Authenticator

Password

Logout

Wagons Pipeline

DRM Dashboard

Component Passports

History & Logs

Spring Analytics

Ask the Manual

System Learning

Shadow Run

Ask the Records

Audit Chain

Export Audit Trail

Shadow run

The app beside the register. The register still governs; nothing the app says releases a wagon during this period. The app's half of every figure below is from its own records; the supervisor writes what it cannot know.

Last 7 days

Export the logs (CSV)

Reading the log

Zero cases where the app passed a spring the register condemned — the one non-negotiable.

DISCREPANCIES

3

app right 2 · register right 1 · both wrong 0 · unresolved 0

WEEK ON WEEK

3

no previous week yet

AMBER BOX ANSWERED

—

0 raised · 0 re-measured · 0 stands · 0 unanswered

ONE WAGON, BOTH WAYS

38 → 31 min

1 paired observation(s) — a large difference is visible, a precise one is not.

1 paired timing(s). Enough to notice a large difference, not enough to defend a precise one — say so rather than report a figure the sample cannot carry.

What the app recorded, day by day

Worked time excludes idle gaps over 15 minutes, and is an upper bound on effort.

DAY	WAGONS SWEEPED	SPRINGS	CONDEMNED	NESTS FLAGGED	BENCH MIN	SPRINGS/H	VERDICTS	AMBER RAISED	ANSWERED	DISCREPANCIES
2026-09-15	1	110	5	0	118	55.9 per hour	40	0	0	0
2026-09-16	1	104	5	0	118	52.9 per hour	39	0	0	0
2026-09-17	1	94	4	0	118	47.8 per hour	41	0	0	1
2026-09-18	1	132	5	0	119	66.6 per hour	23	0	0	1
2026-09-19	3	113	5	0	118	57.5 per hour	22	0	0	1

shadow readonly

DO

1. Shadow Run.

YOU MUST SEE

► The week and the verdict are visible; there are **no record buttons** — the DRM reads, the designated reviewer writes.

WHAT IT PROVES

Oversight sees everything and writes nothing.

CHECKED ON THE BUILD

verified · 02:29:32 pm

- ✓ the week is visible
- ✓ no record buttons for the DRM

IR

WRS Raipur

RDSO G-95 Rev-II

Online

हिंदी

DRM (Divisional Railway Manager)

DRM — DIVISIONAL OFFICER

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Ask the Manual

System Learning

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Ask the Records

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Export Audit Trail

Ask the records

A question about this shop's own records, answered by fixed arithmetic over rows you can see. No model writes a number here.

No model is configured: the matcher alone reads your sentence, offline.

is any gauge reading high

Ask

WHETHER ANY GAUGE READS A MILLIMETRE OR MORE FROM THE SHOP'S OTHER GAUGES ON THE SAME KIND OF SPRING.

question matched offline

OSG-02: Disagrees with OSG-01, the only other gauge on this kind, by 1.5 mm (median 258.5 vs 257, 245 vs 283 readings). With two gauges the record cannot say which one is off — put both against the master. OSG-01: Disagrees with OSG-02, the only other gauge on this kind, by 1.5 mm (median 257 vs 258.5, 283 vs 245 readings). With two gauges the record cannot say which one is off — put both against the master. OSG-02: Disagrees with OSG-01, the only other gauge on this kind, by 1.5 mm (median 258.5 vs 257, 430 vs 433 readings). With two gauges the record cannot say which one is off — put both against the master. OSG-01: Disagrees with OSG-02, the only other gauge on this kind, by 1.5 mm (median 257 vs 258.5, 433 vs 430 readings). With two gauges the record cannot say which one is off — put both against the master.

OSG-02 on CASNUB_22_HS[USED]OUTER
+1.5 mm · n=245

OSG-01 on CASNUB_22_HS[USED]OUTER
-1.5 mm · n=283

OSG-02 on CASNUB_22_NLB[USED]OUTER
+1.5 mm · n=430

OSG-01 on CASNUB_22_NLB[USED]OUTER
-1.5 mm · n=433

GAUGE	KIND	N	MEDIANMM	OTHERSMEDIANMM	SHIFTMM	FLAGGED
OSG-02	CASNUB_22_HS[USED]OUTER	245	258.5	257	1.5	yes
OSG-01	CASNUB_22_HS[USED]OUTER	283	257	258.5	-1.5	yes
OSG-02	CASNUB_22_NLB[USED]OUTER	430	258.5	257	1.5	yes
OSG-01	CASNUB_22_NLB[USED]OUTER	433	257	258.5	-1.5	yes
ISG-01	CASNUB_22_NLB[USED]INNER	695	259.5	—	—	

ask

DO

1. Ask the Records → is any gauge reading high → How this was computed.

YOU MUST SEE

► OSG-02, by how much, and what the record can and cannot say; the query and the rows.

WHAT IT PROVES

As 3.18 — the DRM can ask the same questions with the same proof.

CHECKED ON THE BUILD

verified · 02:29:35 pm

✓ OSG-02 named

✓ how it was computed, with the rows

IR

WRS Raipur

RDSO G-95 Rev-II

Online

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DRM (Divisional Railway Manager)

DRM — DIVISIONAL OFFICER

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Audit Chain

Export Audit Trail

Audit chain

SHA-256 across every event, re-derived from the first entry each time you ask.

Verify chain again

Chain intact

Re-derived just now — at 2:29:47 pm, in 15 ms (check 2 this session)

Audit chain verified — 11 entries, unbroken.

ENTRIES

11

CHECKED AT

21/9/2026, 2:29:47 pm

What is covered

✓ Every spring verdict, including the ones recorded offline and sent later.

✓ Every stage transition, with who moved it and when.

✓ Every release certificate, with its own keyed HMAC over its contents.

✓ Every change of role — a changed role breaks the chain, not just changed data.

Database triggers make the log append-only. The same check runs at GET /api/audit/verify for anyone who would rather not trust a screen.

What a pass does not prove

That no record was altered after it was written is not the same as every measurement having been correct. A wrong reading, honestly entered, hashes exactly as well as a right one. A verified chain says the register is intact. It says nothing about whether the caliper was read properly.

audit

DO

1. Audit Chain → Verify chain again.

YOU MUST SEE

► Chain intact; Re-derived just now — at HH:MM:SS, in N ms.

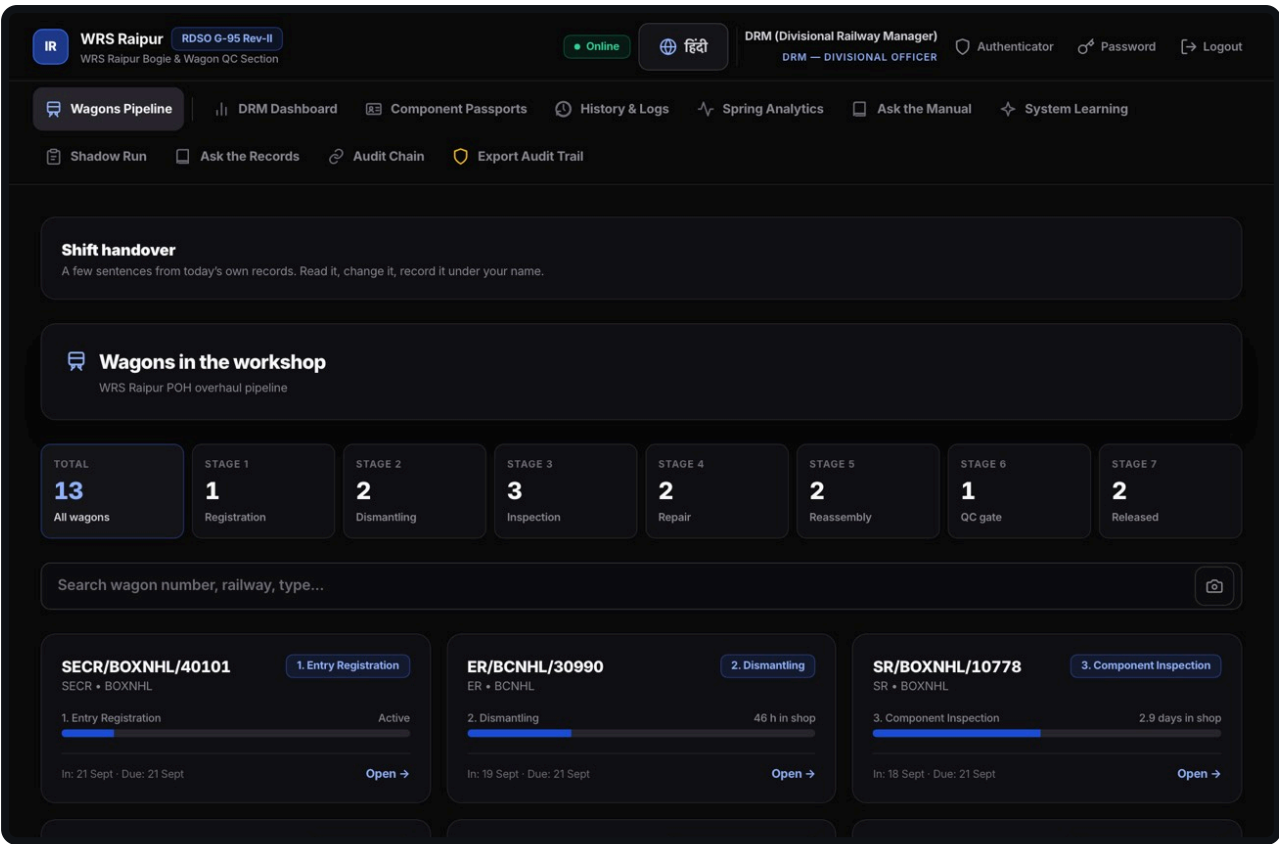
WHAT IT PROVES

The DRM can prove the record unaltered without asking anyone.

CHECKED ON THE BUILD

verified · 02:29:43 pm

✓ "Chain intact Re-derived just now — at 2:29:47 pm, in 15 ms (check 2 this session"



pipeline drm

DO

1. **Wagons Pipeline.** Look for a Register button.

YOU MUST SEE

- There is **no Register New Wagon button** — and the server refuses the DRM's sign-in for it too (403).
- As a supervisor: a junk number is refused; a wagon registered by mistake is voided with *Registered in error?* — reason, one-time code — and leaves the pipeline while its record stays.

WHAT IT PROVES

The record is always written by the floor. Oversight cannot register or inspect, so a figure on the DRM's dashboard is never the DRM's own entry.

CHECKED ON THE BUILD

verified · 02:29:51 pm

- ✓ no Register New Wagon button
- ✓ the server refuses the DRM's token: 403

IF IT IS NOT THERE

- A Register button on the DRM's pipeline: a permission fault — report it.

4.9 A certificate that proves itself

☐ Seen it

Release Certificate — SECR/BOXNHL/10492
Form RDSO/WRS-QC-REL, Bogie & Brake Certified

WAGON REPAIR SHOP (WRS), RAIPUR (C.G.)
रॉलिंग स्टॉक गुणवत्ता प्रमाणपत्र / ROLLING STOCK QUALITY RELEASE CERTIFICATE
CERTIFICATE NO. WRS-QC-REL/2026/08/7328

Wagon Number: SECR/BOXNHL/10492 Wagon Type: BOXNHL
Owning Railway: SECR Intake Date: 2026-08-24
Release Date: 2026-08-30 Workshop TAT: 6.2 Days

1. CASNUB BOGIE 8-CATEGORY RDSO QUALITY CLEARANCE MATRIX

Category	Subsystem Inspection Scope	Standard Ref	Clearance Status
1. Springs	Outer, Inner, Snubber (Bogie 1 & 2)	RDSO 0-95 Table 28-33	ALL CLEARED
2. Wheels & Axles	Wheel profile, tread wear, flange, UST	RDSO 0-99/1 /ND-97	ALL CLEARED
3. Bearings	CTRB cartridge bearings, adapter, seals	RDSO 0-81	ALL CLEARED
4. Brake System	Brake blocks, rigging, S&L, cylinder, DY	RDSO 02-489-03	7/7 CLEARED
5. Couplers & Draft	CBC body, knuckle, high-capacity draft gear	RDSO 48-80-08 / ND-70-80-10	ALL CLEARED
6. Bogie Frame	Side frames, bolsters, centre plate, bearings	RDSO 0-95	5/5 CLEARED
7. Friction Wedges	Shoe wear, vertical force, support fit	RDSO 0-95 / WMA 3.0 §3050	3/3 CLEARED

Start Voice Inspection
No voice command system yet. Please microphone or click simulation chips.

Outer Spring Fit Friction Wedge Condition (steep cracks) Inner Spring Replaced CTRB Bearing Replaced Shoe Brake System Under Last Action Spoken clearly (read by the server)

1. Springs (Outer, Inner, Snubber) 2. Wheels & Axles 3. Bearings (CTRB & Adapters) 4. Brake System & Rigging 5. Couplers & Draft Gear (CBC) 6. Bogie Frame & Bolster 7. Friction Wedges 8. Body & Underframe

Auto-synchronized with Phase 1 Spring Classification records
Phase 1 caliper height measurements and RDSO band classifications are live linked into this checklist.

Read the caliper

1. Springs (Outer, Inner, Snubber) 6 items

Inner Spring (Bogie 1) **MANDATORY** **WRS** **PASS** **FAIL** **CONCERNED** **REPAIRED** **REPLACED** **Measure** **Info**

Inner Spring (Bogie 2) **MANDATORY** **WRS** **PASS** **FAIL** **CONCERNED** **REPAIRED** **REPLACED** **Measure** **Info**

Outer Spring (Bogie 1) **MANDATORY** **WRS** **PASS** **FAIL** **CONCERNED** **REPAIRED** **REPLACED** **Measure** **Info**

Note: Inspected from spring measurement: 201 Inset (BLUE, Table 28)
Action: REPAIRED (Overhauled and re-inspected with PASO)

certificate

Verify a release certificate or a wagon passport
Checked here, in this browser, with no request to any server. Paste the certificate's JSON, or the passport file's text — or open the file.

```
{
  "success": true,
  "data": {
    "certificateNumber": "WRS-QC-REL/2026/08/7328",
    "certificateHash": "a7554d2a38ba20b7f98739205a6874e594ec3e52839c61726ec0ba76629a5",
    "issuedAt": "2026-08-24T14:58:28.503Z",
    "wagon": {
      "type": "BOXNHL",
      "number": "SECR/BOXNHL/10492",
      "entryDate": "2026-08-24",
      "inspection": {
        "category": "RDSO",
        "total": 8,
        "cleared": 8,
        "concerned": 0,
        "failed": 0,
        "pending": 0,
        "verified": "CLEARED"
      },
      "totalInspected": 41,
      "passedCount": 41,
      "category": "RDSO",
      "categoryReason": "All items passed inspection",
      "totalInspected": 41,
      "passedCount": 41,
      "category": "RDSO",
      "categoryReason": "All items passed inspection"
    }
  }
}
```

Choose File | No file chosen **Verify**

VERIFIED — signed by the key with fingerprint 409DB815D0C8652A, and unaltered.

Wagon: SECR/BOXNHL/10492
Certificate: WRS-QC-REL/2026/08/7328
Signed at: 2026-08-30T14:58:28.503Z
Supervisor: WRS-SUP-2019
Key fingerprint: 409DB815D0C8652A
Certificate hash: a7554d2a38ba20b7f98739205a6874e594ec3e52839c61726ec0ba76629a5 (the QR on the printed certificate carries its first 18 characters: a7554d2a38ba20b7f98739205a6874e594ec3e52839c61726ec0ba76629a5)

Signature: valid ECDSA over the signed content
Advisories accepted: [{"id": "parts_not_recorded", "category": "BODY_UNDERFRAME", "partName": "41 positions", "issueType": "PARTS_NOT_ACCOUNTED", "description": "41 expected part positions on this wagon type have nothing recorded against them — the parts record covers 0 of 41, remediationAction": "Record what came off and went back on for these positions, or record that they were not disturbed. An unrecorded position is not evidence that nothing is missing."}]

► The signed content, exactly
► What this proves, and what it does not

verify

DO

1. Open **SECR/BOXNHL/10492** → **Release Certificate**.
2. **Scan the QR with the phone** on the same Wi-Fi.

YOU MUST SEE

- The phone opens the verify page and shows **VERIFIED** — the signature checked against the shop's key in the phone's own browser.
- The same page also takes a certificate *file* with no server at all — what another railway would do.

WHAT IT PROVES

No login, no server, no trust in this laptop: the certificate carries its proof.

CHECKED ON THE BUILD

verified • 02:29:54 pm

- ✓ the QR is a link to the verify page
- ✓ the verify page shows VERIFIED: "Verify a release certificate or a wagon passport Checked here, in this browser, with no request to a..."

By hand: The verify page was opened and checked from the laptop; scanning it needs the phone.

IF IT IS NOT THERE

- The phone opens the app's sign-in screen instead: the phone holds an old copy of the app — open the address once more and it updates.

